BCS-067	PATTERN RECOGNITION	L T P 3 0 0	3 Credits

Course Objectives:

CO1	Understand the concept of a pattern and the basic approach to the development of pattern recognition and machine intelligence algorithms
CO2	Understand the basic methods of feature extraction, feature evaluation, and data mining
CO3	Understand and apply both supervised and unsupervised clasification methods to detect and characterize patterns in real-world data
CO4	Develop prototype pattern recognition algorithms that can be used to study algorithm behavior and performance against real-world multivariate data
CO5	To introduce the fundamental algorithms for pattern recognition

MODULE-I

Basics of pattern recognition, Learning Types: Supervised & Unsupervised, Pattern recognition approaches, Decision Boundary, Feature Vector, Mathematical foundations – Probability Theory (Bayes Theorem), Expectation, Mean and Covariance, Normal distribution, multivariate normal densities. Bayesian Decision Theory, Classifiers, discriminant functions, Normal density (Univariate & Multivariate). Distance Based Classifiers**:** Euclidean Distance, Manhattan Distance and Mahalanobis Distance, Adaptive Decision Boundary.

MODULE-II

Principle Component Analysis, Linear discriminant analysis, Hidden Markov Model – Evaluation (Forward Algorithm), Decoding (HMM Decoding Algorithm), Learning (Forward-Backward Algorithm). Artificial Neural Networks: Biological Neuron, Artificial Neuron, Activation Function, Perceptron, Fixed Increment Perceptron Learning algorithm.

MODULE-III

K-Nearest Neighbour Estimation, Nearest Neighbour Rule, Fuzzy classification - Membership functions, interference in fuzzy logic, fuzzy if - then rules.Unsupervised Learning & Clustering: Criterion functions for clustering, Clustering Techniques: Iterative square - error partition clustering – K means, Fuzzy K - Means agglomerative hierarchical clustering, Cluster validation.

Text Books:

1. Richard O. Duda, Peter E. Hart and David G. Stork, "Pattern Classification", 2nd Edition, John Wiley, 2006.
2. Earl Gose, Richard Johnsonbaugh and Steve Jost, "Pattern Recognition and Image Analysis", PHI.
3. Sergios Theodoridis, Konstantinos Koutroumbas, "Pattern Recognition", 2nd Edition, Elsevier Academic Press.
4. S. Rajsekaran & G.A. Vijayalakshmi Pai, "Neural Networks,Fuzzy Logic and Genetic Algorithm Synthesis and Applications" Prentice Hall of India.
5. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2009.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Understand the process of Software Project Management.
CO2	Conduct project planning activities that accurately forecast project costs.
CO3	Handle tools like MS Project & SVN
CO4	Understand the skills required for managing projects, project teams, and stakeholders
CO5	Understand the process of Software Project Management.